

# AI Lab Pratical5 : Image Sementation

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## Introduction

In this section, we will explore the fundamental concepts in order to capture the object shape by applying several techniques, including Thresholds, k-Mean, Mean Shift and Graph Cut for segmentation. For the practical component, you are provided with a blurred image file, “**birds.png**”, located in the “**test\_image**” folder. The objective of this laboratory exercise is to enhance the given image so that it appears clearer and visually improved.

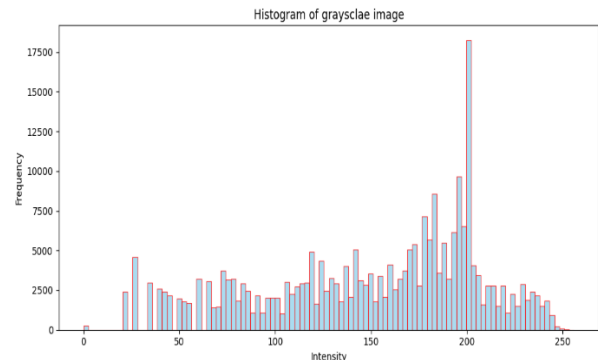


Fig: birds.png

## 1. Binary Image Segmentation



Grayscale Image



Histogram of Image

Defined the intensithy function by  $f(x, y)$  and the thresolds is  $T$ , the binary image is convert the intensity of  $f(x, y)$  into value  $[0,1]$  by:

$$g(x, y) = \begin{cases} 0 & \text{if } f(x, y) \geq T \\ 1 & \text{if } f(x, y) < T \end{cases}$$

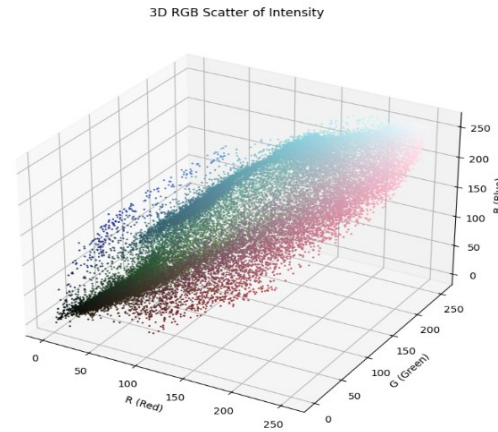
Task: Applied this k-mean technique on “birds.png” image and display your result.

## 2. Sementation as K-mean Clustering

Pixel in Space 3D:



Input Image

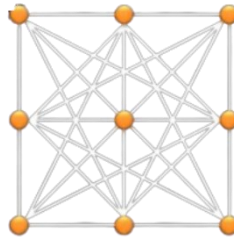


Pixel RGB Color distribution

Task: Applied this k-mean technique on “birds.png” image and display your result.

## 2. Graph Base Segmentation

Let  $i$  and  $j$  be two pixels whose the values of intensity are  $f_i$  and  $f_j$ .



**Pixel Dissimilarity:**

$$d(f_i, f_j) = \sqrt{\sum_k (f_i - f_j)^2}$$

**Pixel Affinity:**

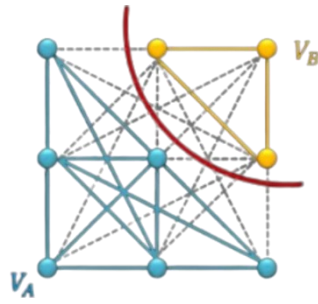
$$w(f_i, f_j) = A(f_i, f_j) = \exp\left(-\frac{1}{2\sigma^2} d(f_i, f_j)\right)$$

Note smaller the **dissimilarity** then it the larger the **affinity**. The objectvie we aim to cut the sub-graph in order to saperated the region of the object.

**Graph Cut:**

**Cut**  $C = (V_A, V_B)$  is a partition of vertices  $V$  of a graph  $G = (V, E)$  into two disjoint subsets  $V_A$  and  $V_B$ .

**Cut-Set:** Set of edges whoes vertices are in different subsets of partition.



**Cost of Cut:** Sum of weights of cut-sets edges.

$$cut(V_A, V_B) = \sum_{u \in V_A, v \in V_B} w(u, v)$$

Objective is ***Min***(***cut***( $V_A, V_B$ ))

Task: Applied this technique on “birds.png” image and display your result.